



A REPORT ON

E-Commerce Coffee Roasters Dashboard

Submitted by:
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Introduction:

The retail and e-commerce industry generates large volumes of transactional data from product sales, customer purchases, and inventory movements. Organizations often struggle to transform this raw data into actionable insights that support strategic decision-making. Business Intelligence dashboards and analytical tools play a crucial role in helping organizations monitor product performance, identify revenue drivers, and optimize product portfolios.

This project presents the development of a **Coffee Roasters Sales Performance Dashboard**, built using Python, Streamlit, SQLite, and Plotly. The dashboard consolidates sales and product performance data into an interactive analytics platform that enables users to evaluate revenue trends, product contribution, category performance, and revenue concentration. The application includes modules such as Product KPI Monitoring, Pareto Analysis, Category Distribution Analysis, ABC Classification, and Product Portfolio Analysis.

From a business perspective, the dashboard demonstrates how data visualization and analytics can improve decision-making by identifying high-performing products, understanding customer purchasing patterns, and highlighting products that contribute most significantly to overall revenue. The project transforms complex transactional data into intuitive visualizations that support both operational monitoring and strategic planning.

Objectives:

The primary objective of this project is to design and implement an interactive dashboard that supports sales monitoring and product performance analysis within a retail environment.

The specific objectives are:

1. Provide a comprehensive overview of business performance through key indicators such as Total Revenue, Units Sold, Product Count, and Average Revenue.
2. Identify top-performing products and categories based on revenue contribution.
3. Perform Pareto Analysis to determine which products generate the majority of total revenue.
4. Classify products using ABC Analysis to support inventory and product management decisions.

5. Analyze category-wise performance and revenue distribution across product groups.
 6. Evaluate product portfolio performance using revenue and sales volume metrics.
 7. Develop a scalable analytical platform using SQLite, Python, Streamlit, and Plotly for future business intelligence applications.
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About the Dataset

The dataset used in this project consists of transactional sales records from a coffee retail business. Each transaction contains information about products sold, sales quantity, pricing, store information, and product classifications.

The dataset is stored in a SQLite database and organized into analytical views that support dashboard reporting and visualization.

Key Data Components

1. Product KPI View (vw_product_kpi)

- Contains product-level sales performance metrics.
- Includes total revenue, units sold, revenue contribution percentage, product category, product type, and product details.
- Supports KPI monitoring and product performance analysis.

2. Category KPI View (vw_category_kpi)

- Aggregates product performance at the category level.
- Includes category revenue, units sold, and category revenue contribution percentages.
- Used for category distribution and comparative analysis.

3. Product Pareto View (vw_product_pareto)

- Contains ranked products based on total revenue.
- Calculates cumulative revenue percentages and identifies products contributing to the top 80% of revenue.
- Supports Pareto and revenue concentration analysis.

4. Transaction Table

- Stores detailed transaction-level information including:
 - Transaction ID
 - Product ID
 - Product Category
 - Product Type
 - Quantity Sold
 - Unit Price
 - Revenue
 - Store Information
 - Transaction Date and Time
-

Relevance of the Dataset to This Project

- Product-level data enables detailed performance monitoring and product ranking.
 - Category-level aggregation supports strategic decision-making regarding product groups.
 - Revenue concentration metrics help identify critical products driving business growth.
 - Transaction-level records provide the foundation for advanced analytical techniques such as ABC Analysis and Product Portfolio Analysis.
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Methodology

1. Data Preparation

The first stage involved loading transactional sales data into a SQLite database. Revenue was calculated by multiplying transaction quantity by unit price.

The data was validated to ensure completeness and consistency before analytical processing. Missing or invalid product records were removed to improve data quality.

The cleaned data was then organized into analytical views that support dashboard reporting.

2. Data Modelling and Integration

SQLite was used as the centralized data repository.

Several SQL views were created to support different analytical requirements:

- Product KPI View
- Category KPI View
- Product Pareto View

These views served as reusable data layers that simplified dashboard development and reduced computational overhead during visualization.

3. Dashboard Design

The dashboard was designed to provide both executive-level insights and detailed analytical capabilities.

The application was divided into three major modules:

- **Product Performance Dashboard**

Provides overall business KPIs and product-level insights.

- **Pareto and Category Analysis**

Highlights revenue concentration patterns and category contributions.

- **Advanced Insights Dashboard**

Provides ABC Analysis and Product Portfolio segmentation.

Interactive filters allow users to dynamically analyze products by category, product type, and ranking thresholds.

4. Implementation Using Streamlit and Python

The dashboard was implemented using Python and Streamlit.

SQLite data was imported into Pandas DataFrames and processed dynamically based on user selections.

Plotly was used to create interactive visualizations including:

- Revenue Bar Charts
- Scatter Plots
- Pareto Curves
- Category Distribution Charts
- Portfolio Matrices
- Revenue Concentration Curves

The application leverages Streamlit components such as:

- KPI Cards
- Tabs
- Sliders
- Select Boxes
- Interactive Data Tables

This architecture enables real-time analysis without requiring complex front-end development.

5. Visual Design and Presentation Logic

The visualization layer was designed to maximize clarity and interpretability.

KPI Cards

Used to display:

- Total Revenue
- Units Sold

- Product Count
- Average Revenue

Bar Charts

Used to compare product and category performance.

Scatter Plots

Used to visualize relationships between revenue and product popularity.

Pareto Charts

Used to identify products responsible for the majority of business revenue.

Portfolio Matrix

Used to classify products based on revenue and sales volume.

The dashboard follows a clean business-oriented design with interactive filters and responsive visualizations.

6. Testing and Validation

Testing was performed to ensure:

- Correct revenue calculations.
- Accurate Pareto ranking.
- Proper ABC classification.
- Dynamic filtering functionality.
- Consistent dashboard performance across different product categories.

Cross-validation was conducted between SQLite outputs and dashboard visualizations to ensure data consistency.

Features of the Operation Dashboard

1. KPI Monitoring

Displays:

- Total Revenue
- Units Sold
- Product Count
- Average Revenue

These metrics provide an immediate overview of business performance.

2. Top Revenue Products

Displays the highest-performing products based on revenue generation.

Users can filter products by category and product type.

3. Product Popularity vs Revenue Analysis

A scatter plot compares:

- Units Sold
- Total Revenue

This helps identify products that generate high revenue despite lower sales volume and vice versa.

4. Pareto Analysis

The Pareto dashboard identifies products contributing to approximately 80% of total business revenue.

Key metrics include:

- Number of products generating 80% of revenue.
- Revenue coverage percentage.
- Product rankings.

This helps management focus on the most valuable products.

5. Category Distribution Analysis

Provides category-level revenue comparison.

Users can identify:

- Best-performing categories.
- Revenue contribution percentages.
- Category sales distribution.

6. ABC Product Analysis

Products are classified into:

A-Class

Products generating approximately 80% of cumulative revenue.

B-Class

Products generating the next 15% of revenue.

C-Class

Products contributing the remaining revenue.

This classification supports inventory prioritization and business planning.

7. Product Portfolio Matrix

Products are segmented into four business categories:

Star Products

High Revenue + High Sales Volume

Premium Products

High Revenue + Low Sales Volume

Volume Drivers

Low Revenue + High Sales Volume

Underperformers

Low Revenue + Low Sales Volume

This analysis supports product portfolio optimization.

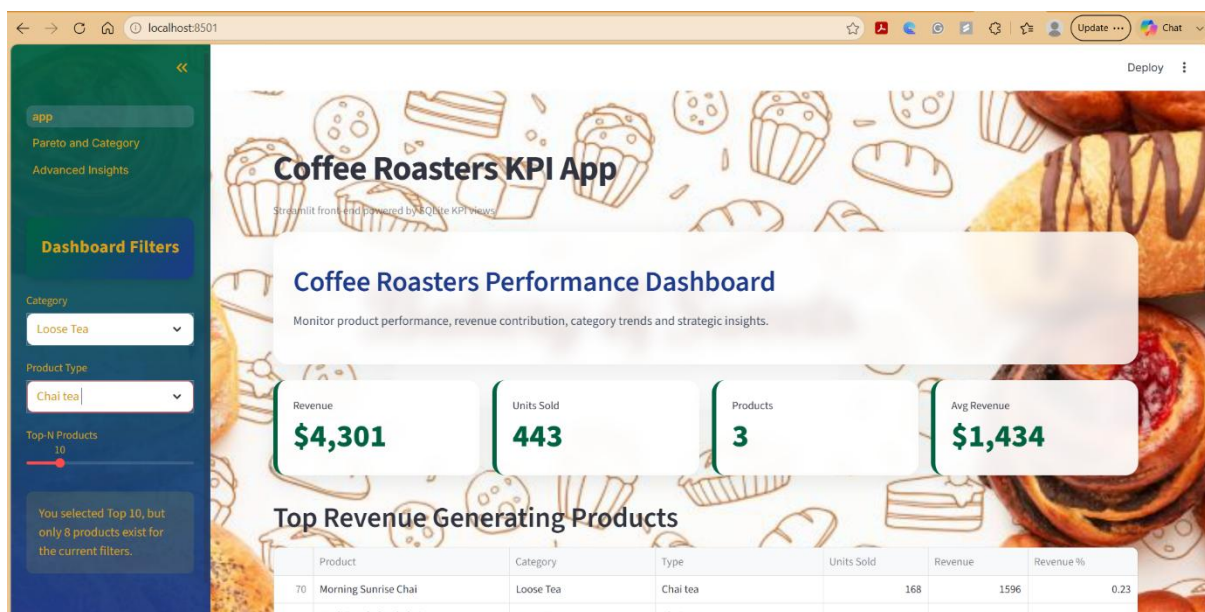
8. Revenue Concentration Analysis

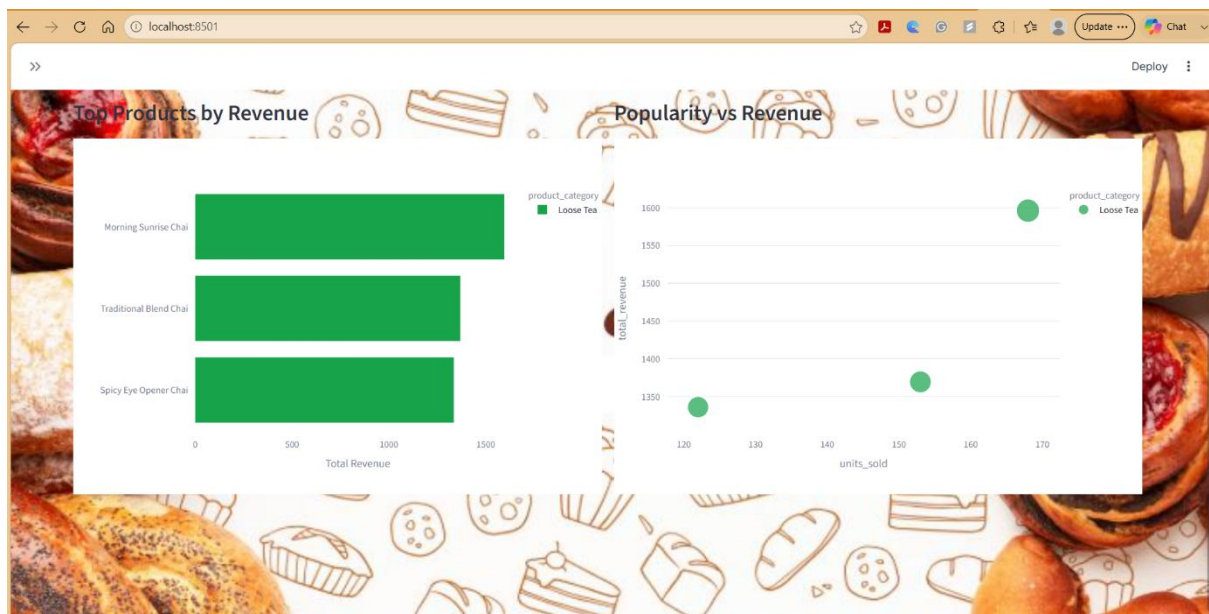
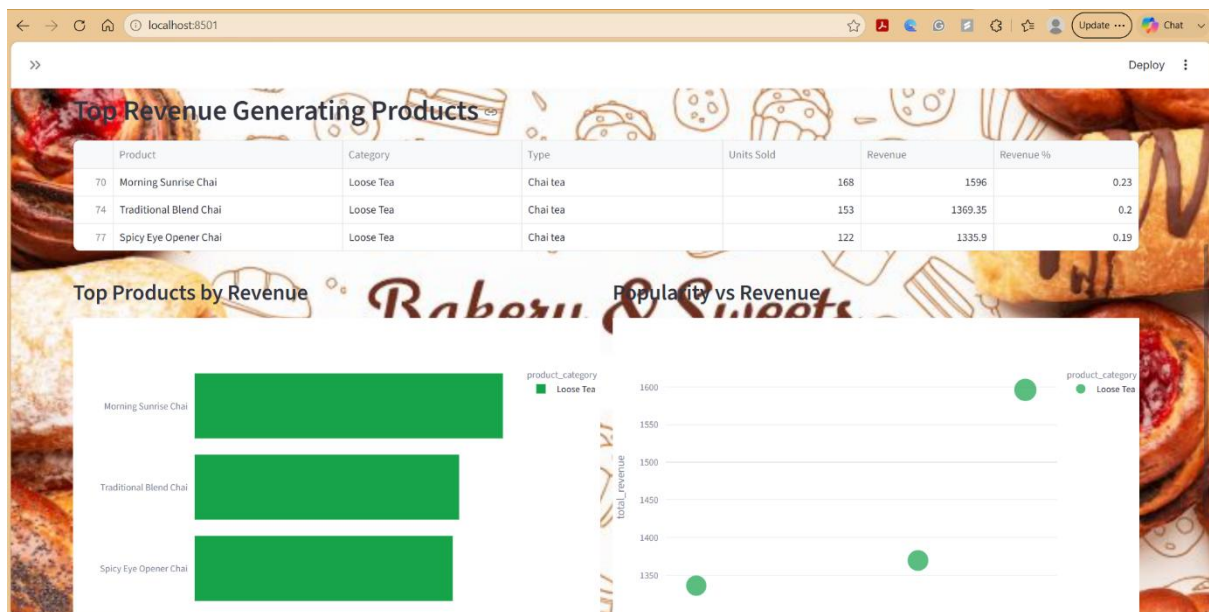
Measures the degree to which business revenue depends on a small number of products.

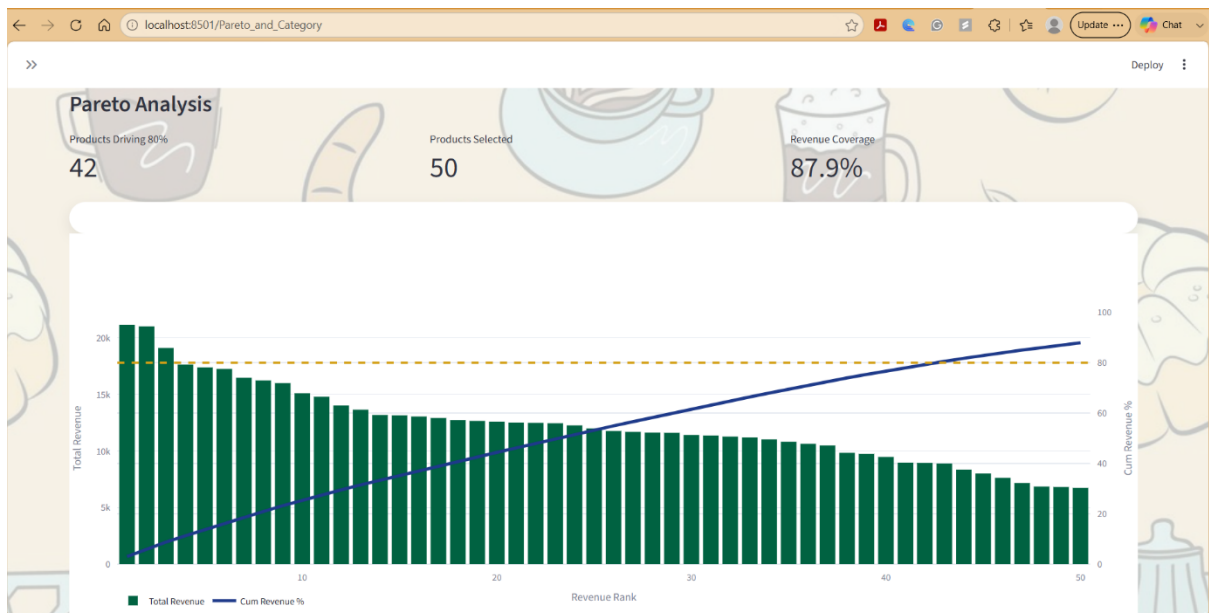
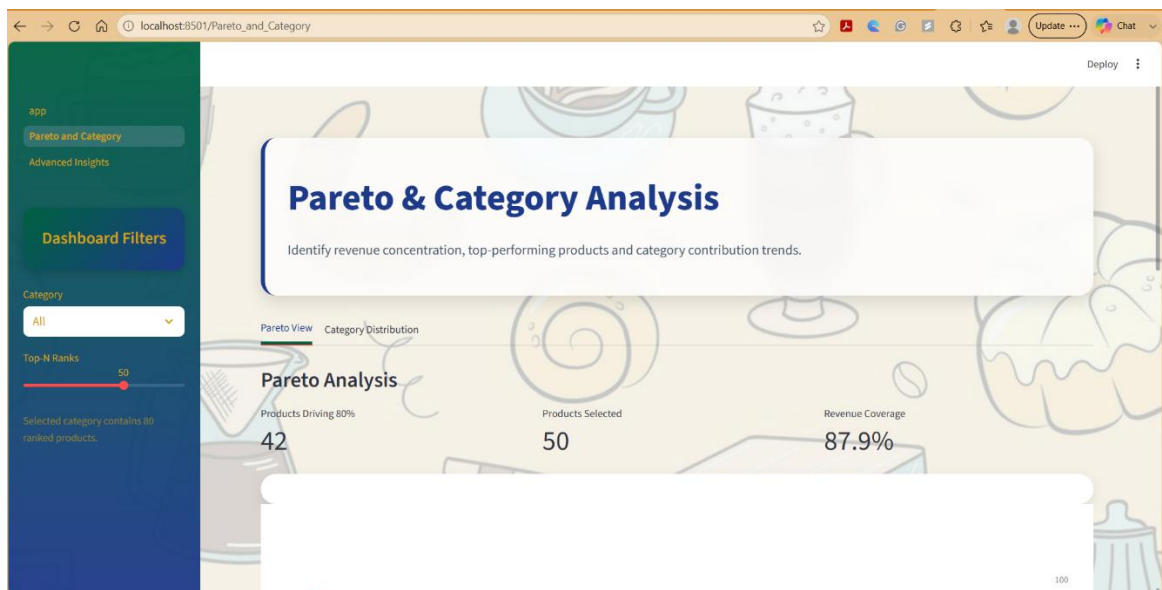
The dashboard identifies:

- Products driving 80% of revenue.
- Top 10 revenue-generating products.
- Revenue concentration trends.

Images of the dashboard:



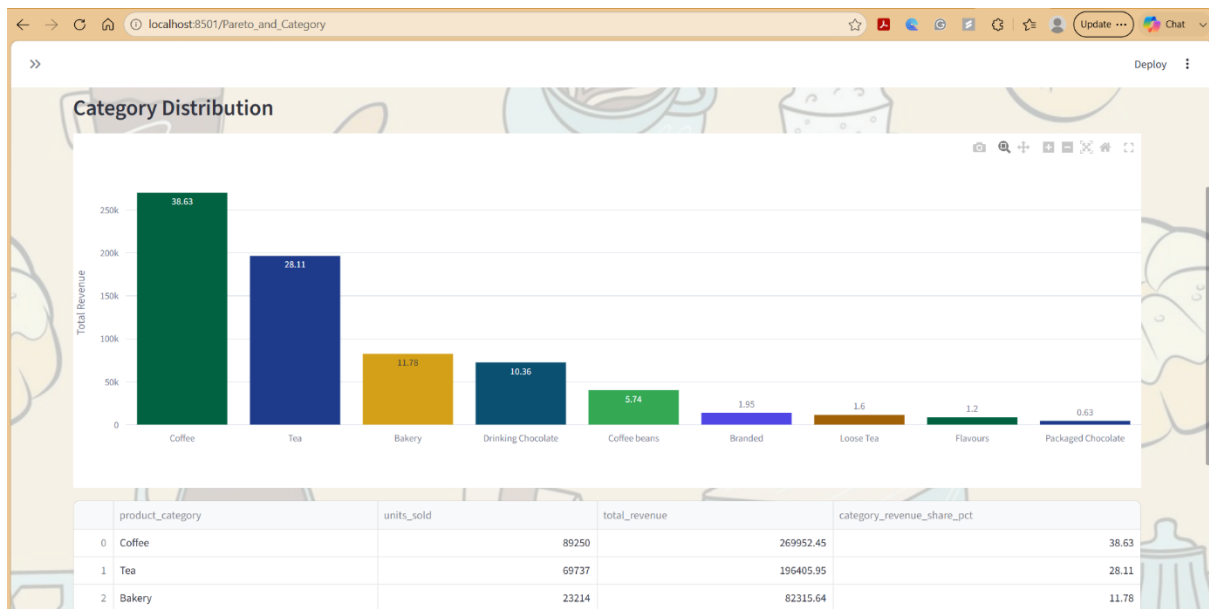
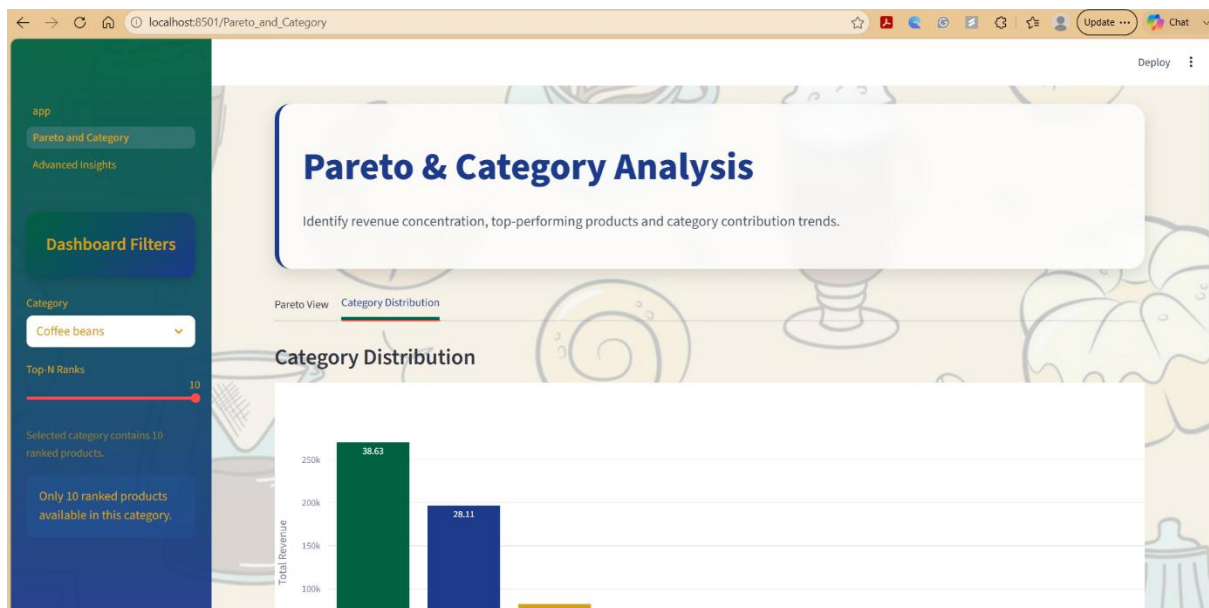




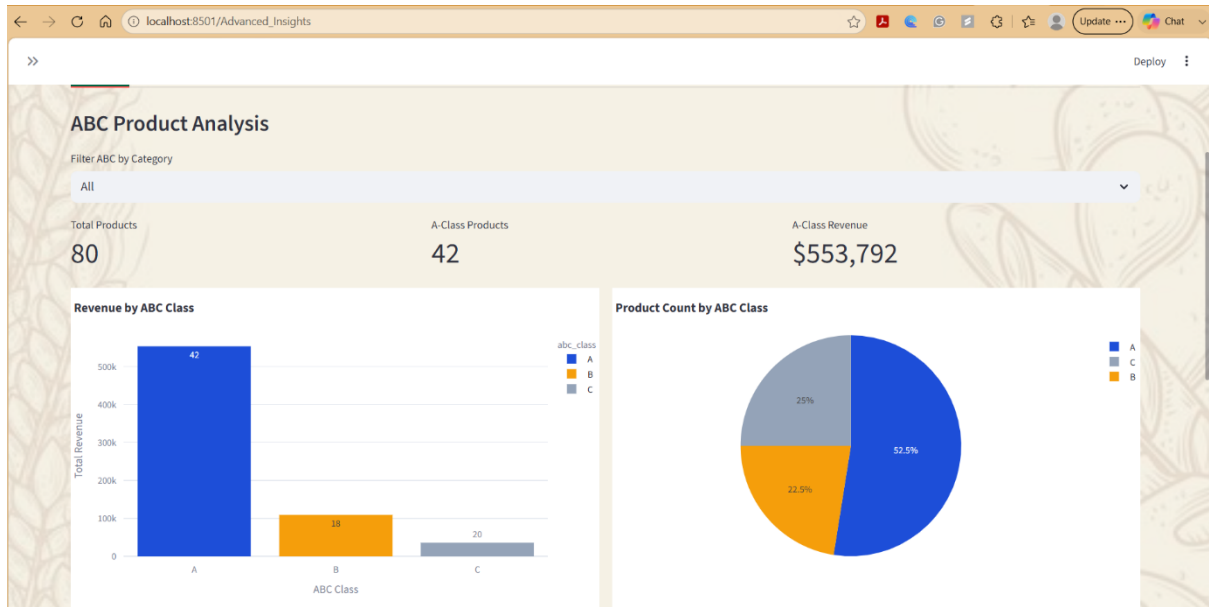
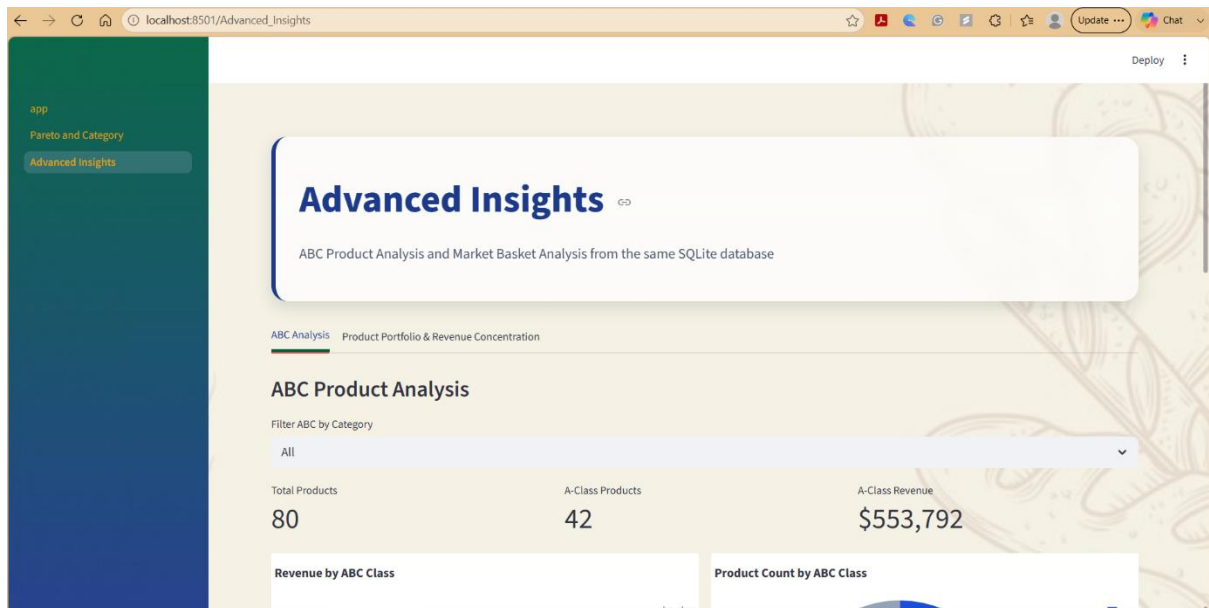
localhost:8501/Pareto_and_Category

Products Contributing to Top 80% of Revenue

	product_detail	product_category	revenue_rank	total_revenue	revenue_share_pct	cum_revenue_pct
0	Sustainably Grown Organic Lg	Drinking Chocolate	1	21151.75	3.03	3.03
1	Dark chocolate Lg	Drinking Chocolate	2	21006	3.01	6.03
2	Latte Rg	Coffee	3	19112.25	2.73	8.77
3	Cappuccino Lg	Coffee	4	17641.75	2.52	11.29
4	Morning Sunrise Chai Lg	Tea	5	17384	2.49	13.78
5	Latte	Coffee	6	17257.5	2.47	16.25
6	Jamaican Coffee River Lg	Coffee	7	16481.25	2.36	18.61
7	Sustainably Grown Organic Rg	Drinking Chocolate	8	16233.75	2.32	20.93
8	Cappuccino	Coffee	9	15997.5	2.29	23.22
9	Brazilian Lg	Coffee	10	15109.5	2.16	25.38



	product_category	units_sold	total_revenue	category_revenue_share_pct
0	Coffee	89250	269952.45	38.63
1	Tea	69737	196405.95	28.11
2	Bakery	23214	82315.64	11.78
3	Drinking Chocolate	17457	72416	10.36
4	Coffee beans	1828	40085.25	5.74
5	Branded	776	13607	1.95
6	Loose Tea	1210	11213.6	1.6
7	Flavours	10511	8408.8	1.2
8	Packaged Chocolate	487	4407.64	0.63



localhost:8501/Advanced_Insights

Deploy

ABC Product Table

	Product	Category	Type	Revenue	Revenue %	Cumulative Revenue %	ABC Class
35	Spicy Eye opener Cnai Rg	Tea	Brewed Cnai tea	\$10,536	1.52	71.02	A
36	English Breakfast Rg	Tea	Brewed Black tea	\$10,500	1.50	72.52	A
37	Jamaican Coffee River Sm	Coffee	Premium brewed coffee	\$9,844	1.41	73.93	A
38	Ethiopia Sm	Coffee	Gourmet brewed coffee	\$9,753	1.40	75.33	A
39	Brazilian Sm	Coffee	Organic brewed coffee	\$9,482	1.36	76.68	A
40	Our Old Time Diner Blend Sm	Coffee	Drip coffee	\$8,968	1.28	77.97	A
41	Scottish Cream Scone	Bakery	Scone	\$8,949	1.28	79.25	A
42	Ouro Brasileiro shot	Coffee	Barista Espresso	\$8,902	1.27	80.52	B
43	Columbian Medium Roast Sm	Coffee	Gourmet brewed coffee	\$8,356	1.20	81.72	B
44	Ginger Scone	Bakery	Scone	\$8,012	1.15	82.86	B
45	Jumbo Savory Scone	Bakery	Scone	\$7,627	1.09	83.96	B

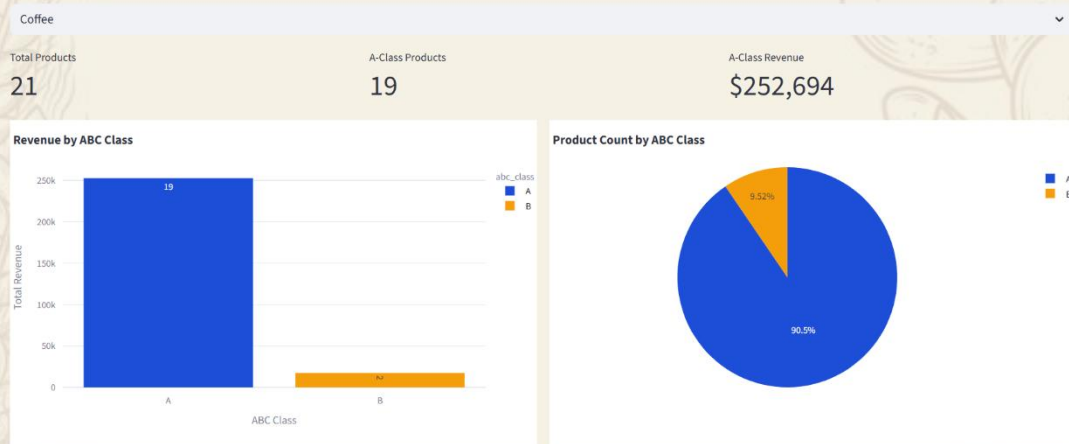
A = top revenue-driving products up to 80% cumulative revenue, B = next 15%, C = remaining products.

	Product	Category	Type	Revenue	Revenue %	Cumulative Revenue %	ABC Class
56	I Need My Bean! Latte cup	Branded	Housewares	\$4,509	0.65	93.40	B
57	Brazilian - Organic	Coffee beans	Organic Beans	\$3,852	0.55	93.95	B
58	Our Old Time Diner Blend	Coffee beans	House blend Beans	\$3,294	0.47	94.42	B
59	Primo Espresso Roast	Coffee beans	Espresso Beans	\$3,068	0.44	94.86	B
60	I Need My Bean! Diner mug	Branded	Housewares	\$2,935	0.42	95.28	C
61	Jamaican Coffee River	Coffee beans	Premium Beans	\$2,884	0.41	95.70	C
62	Espresso Roast	Coffee beans	Espresso Beans	\$2,493	0.36	96.05	C
63	Sugar Free Vanilla syrup	Flavours	Sugar free syrup	\$2,324	0.33	96.38	C
64	Columbian Medium Roast	Coffee beans	Gourmet Beans	\$2,220	0.32	96.70	C
65	Chocolate syrup	Flavours	Regular syrup	\$2,126	0.30	97.01	C
66	Caramel syrup	Flavours	Regular syrup	\$2,061	0.30	97.30	C

A = top revenue-driving products up to 80% cumulative revenue, B = next 15%, C = remaining products.

	Product	Category	Type	Revenue	Revenue %	Cumulative Revenue %	ABC Class
70	Morning Sunrise Chai	Loose Tea	Chai tea	\$1,596	0.23	98.32	C
71	Serenity Green Tea	Loose Tea	Green tea	\$1,471	0.21	98.53	C
72	English Breakfast	Loose Tea	Black tea	\$1,441	0.21	98.74	C
73	Peppermint	Loose Tea	Herbal tea	\$1,369	0.20	99.13	C
74	Traditional Blend Chai	Loose Tea	Chai tea	\$1,369	0.20	99.13	C
75	Lemon Grass	Loose Tea	Herbal tea	\$1,360	0.19	99.33	C
76	Guatemalan Sustainably Grown	Coffee beans	Green beans	\$1,340	0.19	99.52	C
77	Spicy Eye Opener Chai	Loose Tea	Chai tea	\$1,336	0.19	99.71	C
78	Earl Grey	Loose Tea	Black tea	\$1,271	0.18	99.89	C
79	Dark chocolate	Packaged Chocolate	Drinking Chocolate	\$755	0.11	100.00	C

A = top revenue-driving products up to 80% cumulative revenue, B = next 15%, C = remaining products.



ABC Product Table

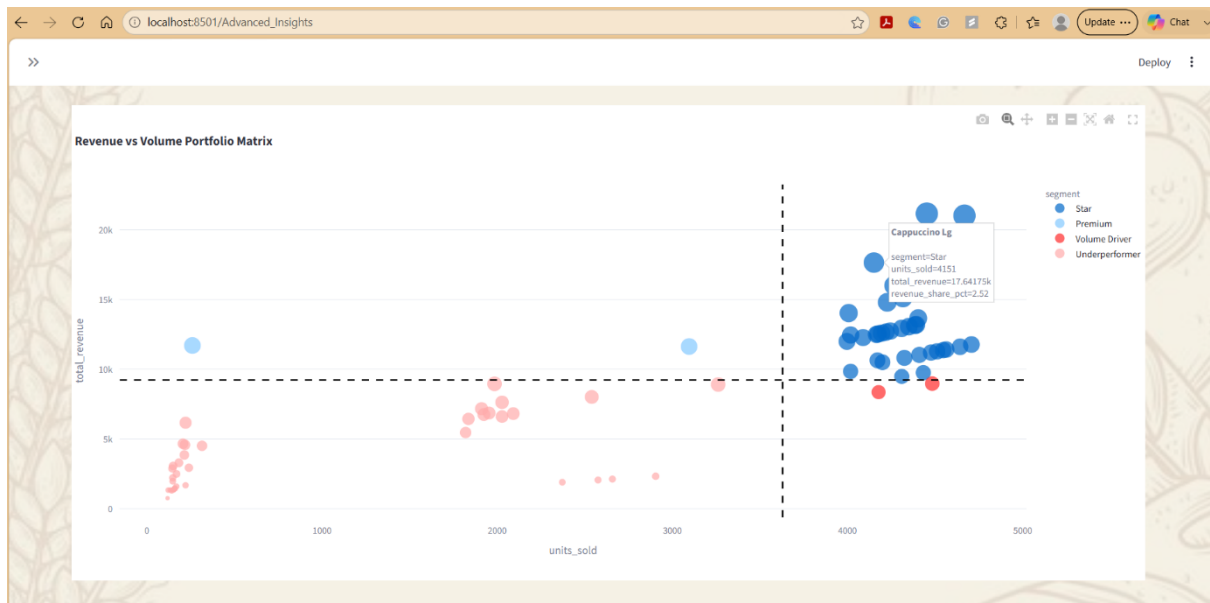
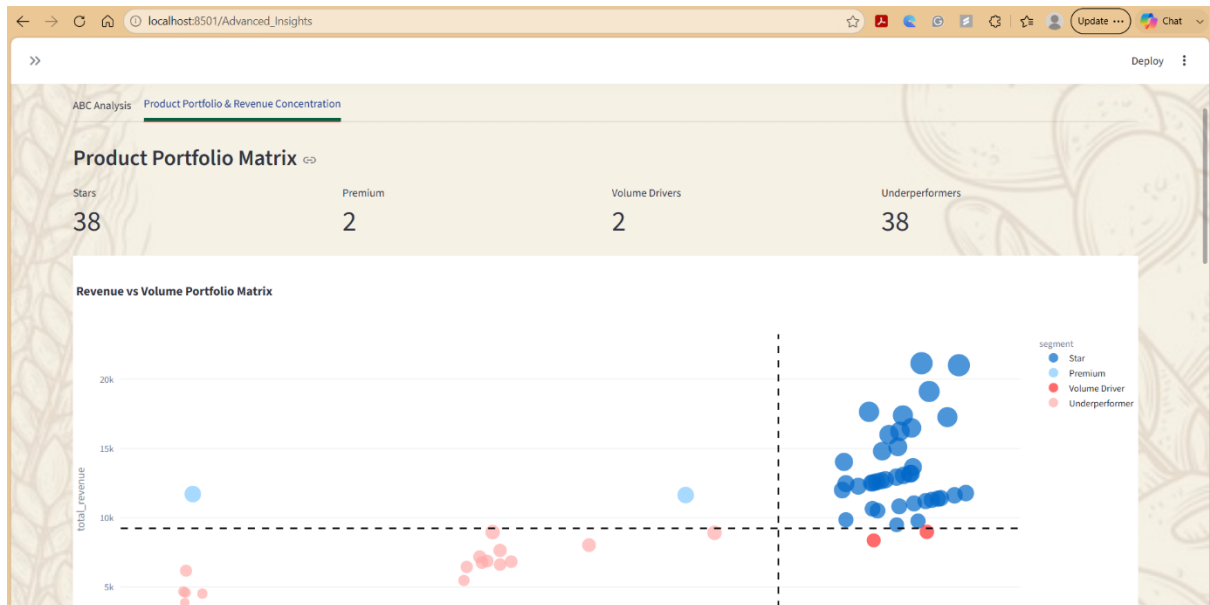
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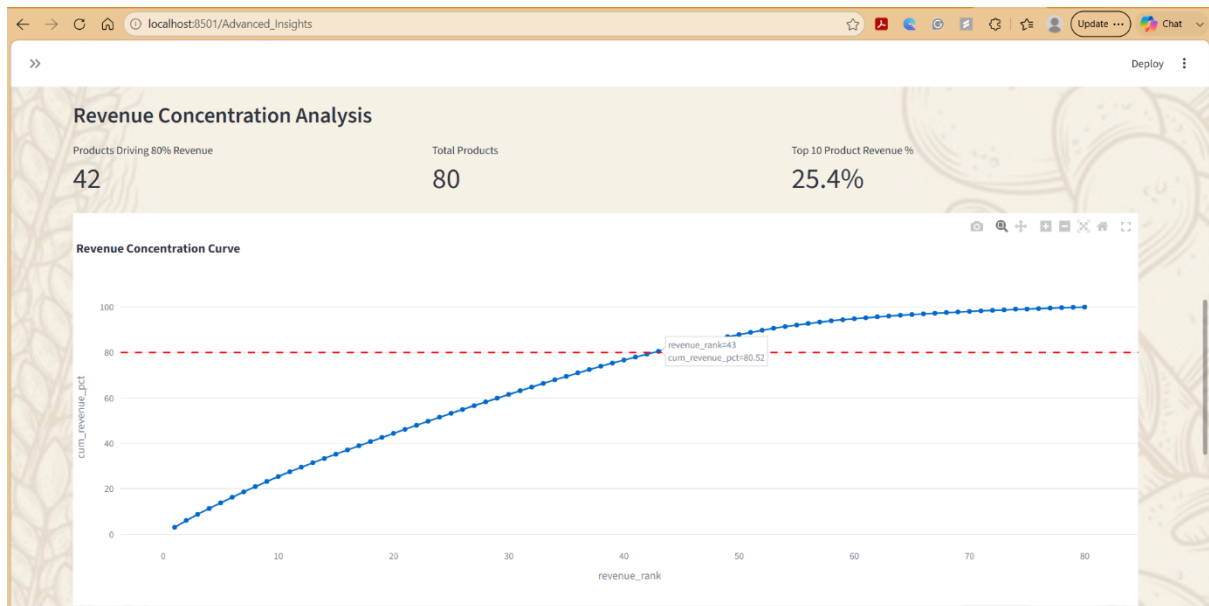
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ABC Product Table

	Product	Category	Type	Revenue	Revenue %	Cumulative Revenue %	ABC Class
22	Jamaican Coffee River Rg	Coffee	Premium brewed coffee	\$12,456	1.78	49.74	A
24	Our Old Time Diner Blend Lg	Coffee	Drip coffee	\$11,991	1.72	53.21	A
30	Columbian Medium Roast Rg	Coffee	Gourmet brewed coffee	\$11,368	1.63	63.16	A
33	Our Old Time Diner Blend Rg	Coffee	Drip coffee	\$11,025	1.58	67.95	A
37	Jamaican Coffee River Sm	Coffee	Premium brewed coffee	\$9,844	1.41	73.93	A
38	Ethiopia Sm	Coffee	Gourmet brewed coffee	\$9,753	1.40	75.33	A
39	Brazilian Sm	Coffee	Organic brewed coffee	\$9,482	1.36	76.68	A
40	Our Old Time Diner Blend Sm	Coffee	Drip coffee	\$8,968	1.28	77.97	A
42	Ouro Brasileiro shot	Coffee	Barista Espresso	\$8,902	1.27	80.52	B
43	Columbian Medium Roast Sm	Coffee	Gourmet brewed coffee	\$8,356	1.20	81.72	B

A = top revenue-driving products up to 80% cumulative revenue, B = next 15%, C = remaining products.





Products Driving 80% Revenue

	Product	Category	Revenue	Cumulative Revenue %
0	Sustainably Grown Organic Lg	Drinking Chocolate	21151.75	3.03
1	Dark chocolate Lg	Drinking Chocolate	21006	6.03
2	Latte Rg	Coffee	19112.25	8.77
3	Cappuccino Lg	Coffee	17641.75	11.29
4	Morning Sunrise Chai Lg	Tea	17384	13.78
5	Latte	Coffee	17257.5	16.25
6	Jamaican Coffee River Lg	Coffee	16481.25	18.61
7	Sustainably Grown Organic Rg	Drinking Chocolate	16233.75	20.93
8	Cappuccino	Coffee	15997.5	23.22
9	Brazilian Lg	Coffee	15109.5	25.38

42 out of 80 products generate approximately 80% of total revenue.

Tableau Dashboard

Dashboard 1: Product Performance and Revenue Contribution

This dashboard focuses on identifying high-performing products, understanding revenue concentration, and evaluating the relationship between product popularity and revenue generation.

Popularity vs Revenue Scatter Plot

The scatter plot compares **Units Sold** against **Total Revenue** for individual products across different product categories.

Key observations:

- Coffee products occupy the upper-right section of the chart, indicating both high sales volume and high revenue generation.
- Products such as **Barista Espresso**, **Gourmet Brewed Coffee**, and **Premium Brewed Coffee** demonstrate strong business performance.
- Bakery and packaged products generally generate lower revenue and sales volumes compared to coffee and tea products.
- The visualization helps identify products that are highly profitable despite lower sales volume and products that rely heavily on sales volume for revenue generation.

Pareto Analysis: Revenue Concentration by Product Rank

The Pareto chart ranks products based on revenue contribution and displays cumulative revenue percentage.

Key findings:

- A relatively small group of products contributes the majority of overall revenue.
- The cumulative revenue curve crosses the 80% threshold after a limited number of products, confirming the Pareto Principle (80/20 Rule).
- Revenue declines gradually as product rank increases, indicating that top-performing products play a significant role in business profitability.

- Management can use this insight to prioritize inventory, promotions, and pricing strategies for the highest-contributing products.

Top Products by Units Sold

The treemap visual highlights products with the highest sales volumes.

Key findings:

- Coffee and Tea products dominate sales volume across the business.
- **Gourmet Brewed Coffee, Barista Espresso, and Brewed Chai Tea** are among the highest-selling products.
- Larger rectangles represent products with higher units sold, while color intensity reflects sales magnitude.
- The visualization helps identify customer preferences and products driving transaction volume.

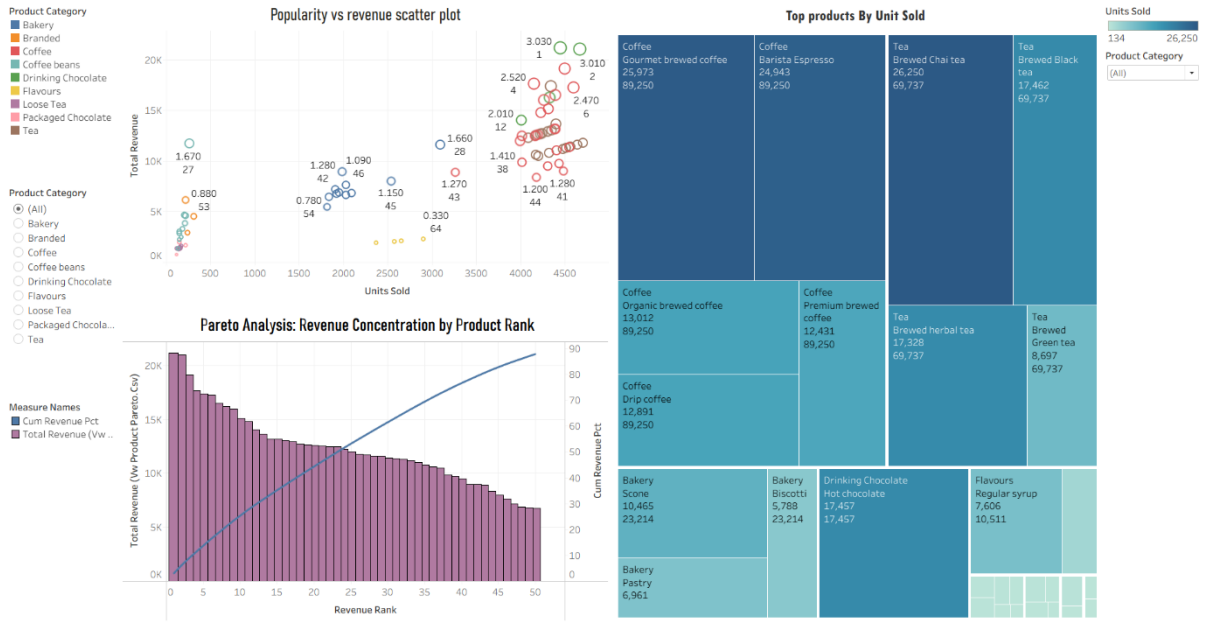
Interactive Filtering

The dashboard allows users to filter by product category.

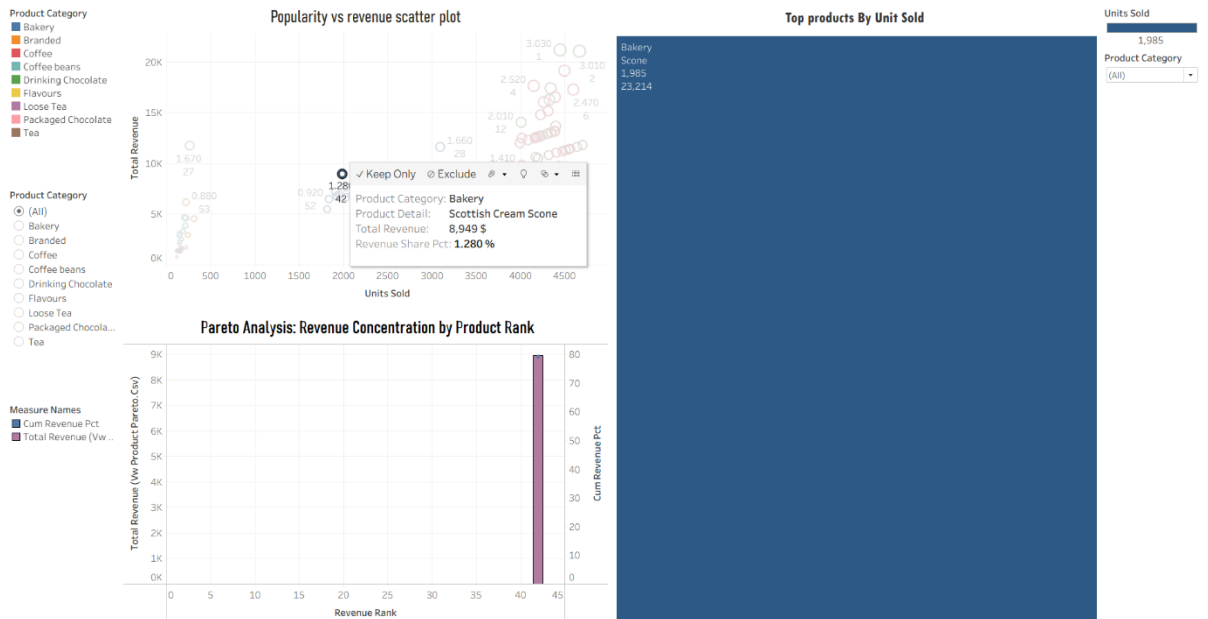
For example:

- Selecting **Bakery** isolates bakery products and reveals that items such as **Scones** and **Pastries** contribute the majority of bakery revenue.
- Selecting **Coffee** highlights the strong revenue concentration among premium coffee products.
- Dynamic filtering enables detailed category-level analysis and performance comparison.

Product Performance and Revenue Contribution



Product Performance and Revenue Contribution



Dashboard 2: Revenue Mix and Category Leaders

This dashboard focuses on understanding how revenue is distributed across categories and identifying category-leading products.

Product Revenue Flow by Category and Product Detail

The Sankey diagram visualizes the relationship between individual products and their corresponding categories.

Key observations:

- Coffee products contribute the largest share of total revenue among all categories.
- Multiple high-performing products flow into the Coffee category, indicating strong category dominance.
- Tea represents the second-largest revenue contributor with several high-performing products.
- Smaller categories such as Flavours, Loose Tea, and Packaged Chocolate contribute comparatively less revenue.

The Sankey visualization provides a clear understanding of how individual products collectively contribute to category-level performance.

Top Products by Revenue and Category

The packed bubble chart displays top products based on revenue contribution.

Key findings:

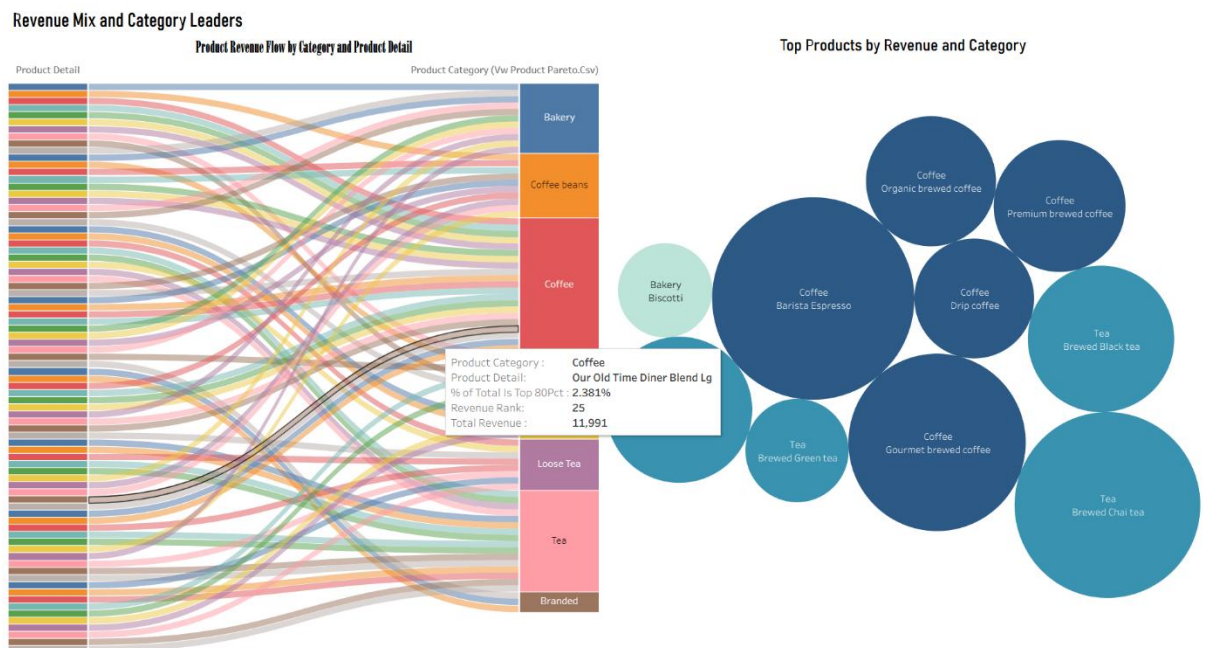
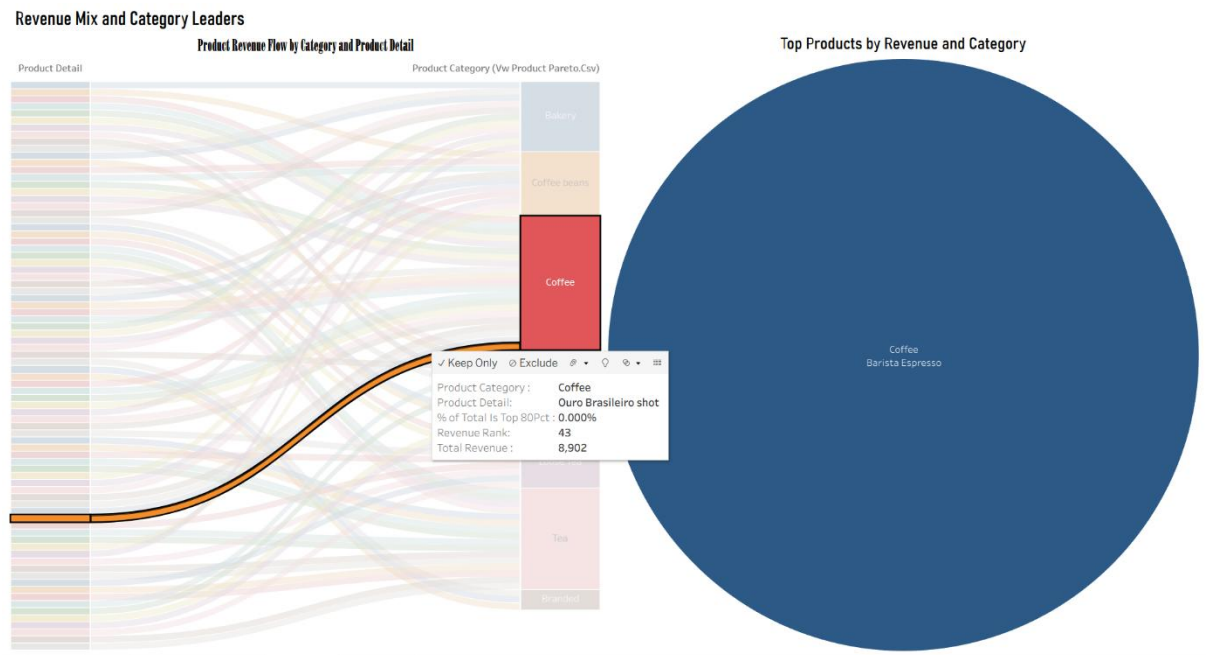
- **Barista Espresso** appears as the largest bubble, indicating the highest revenue contribution.
- Other major revenue-generating products include:
 - Gourmet Brewed Coffee
 - Premium Brewed Coffee
 - Organic Brewed Coffee
 - Brewed Chai Tea
 - Brewed Black Tea
- Coffee products dominate the top revenue rankings, demonstrating strong customer demand and profitability.
- Tea products also contribute significantly but remain secondary to coffee products.

Interactive Drill-Down Analysis

The dashboard supports interactive exploration through category selection.

Examples include:

- Selecting the Coffee category filters the bubble chart to show only top-performing coffee products.
- Selecting Flavours highlights flavour-related products and their revenue contribution.
- Users can quickly identify category leaders and compare revenue distribution across categories.



Dashboard 3: Beverage Revenue and Menu Breadth

This dashboard evaluates product assortment breadth and category-level product performance.

Top-80 Products Table

The table displays products that collectively contribute approximately 80% of total business revenue.

Key findings:

- A limited number of products generate the majority of revenue.
- Premium coffee beverages dominate the top-performing product list.
- Revenue percentages help prioritize products for marketing and inventory decisions.

Beverage Menu Breadth and Volume by Category

The heatmap visualizes product variety and sales volume across beverage categories.

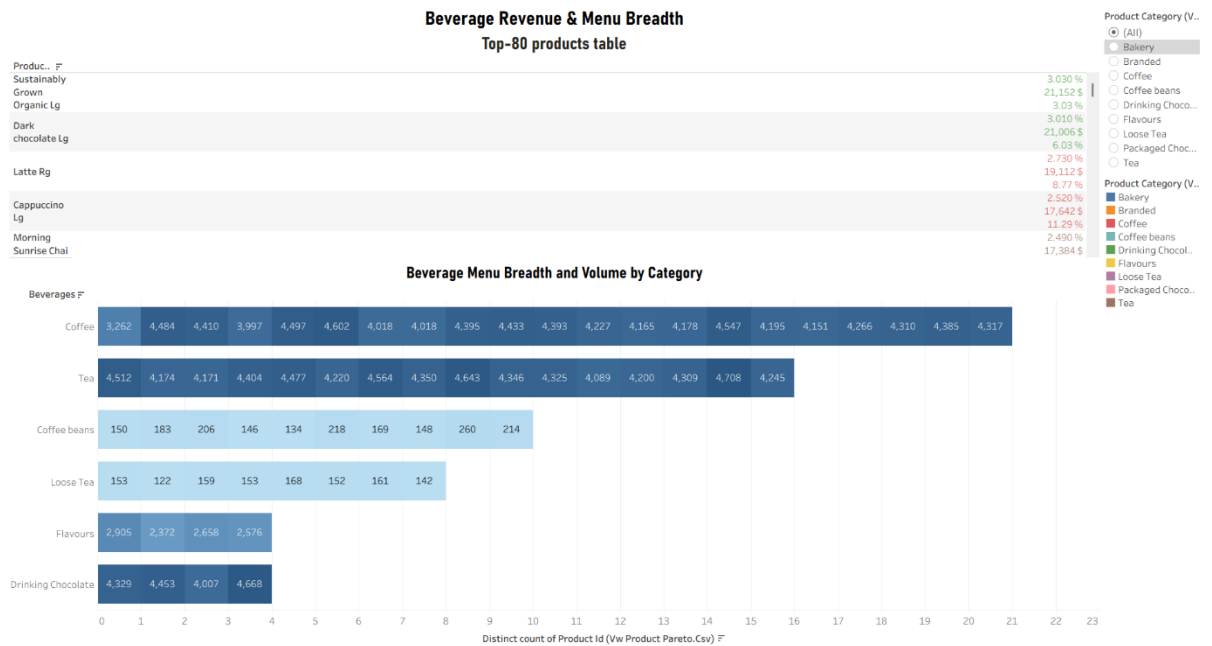
Key observations:

- Coffee has the largest number of products and the highest sales volume.
- Tea also maintains a broad product portfolio with strong sales performance.
- Categories such as Coffee Beans and Loose Tea contain fewer products and lower overall sales volumes.
- Drinking Chocolate and Flavours contribute moderate revenue but have smaller product portfolios.

Business Insights

The analysis demonstrates that:

- Coffee is the dominant category in terms of both menu breadth and revenue generation.
- Tea serves as a strong secondary category supporting overall sales.
- Smaller categories provide diversification but contribute less significantly to overall revenue.
- Maintaining a balanced portfolio between high-volume categories and niche products can support sustainable revenue growth.



Overall Insights from Tableau Dashboards

The combined Tableau dashboards reveal several important business insights:

- Coffee products are the primary drivers of both revenue and sales volume.
- A small number of products contribute approximately 80% of total business revenue.
- Barista Espresso, Gourmet Brewed Coffee, Premium Brewed Coffee, and Brewed Chai Tea consistently emerge as top-performing products.
- Revenue is highly concentrated among a limited set of products, confirming the Pareto principle.
- Coffee and Tea categories dominate the product portfolio and should remain strategic focus areas.
- Interactive filtering and drill-down capabilities enable managers to investigate category-specific performance and identify opportunities for product optimization.

These visualizations provide decision-makers with a comprehensive understanding of product performance, revenue concentration, category contribution, and portfolio effectiveness, enabling more informed business planning and inventory management.

Results and Insights:

Several business insights emerged from the dashboard analysis:

- A relatively small percentage of products generate the majority of total revenue, confirming the Pareto principle.
- A-Class products contribute the highest share of business revenue and should receive priority attention.
- Certain categories consistently outperform others in terms of revenue contribution.
- Product Portfolio Analysis reveals opportunities for promoting Volume Drivers and improving Underperformers.
- Revenue concentration analysis highlights key products that have a significant impact on overall business performance.

These findings demonstrate the value of analytical dashboards in supporting data-driven retail decision-making.

Conclusion and Future Scope

The Coffee Roasters Performance Dashboard successfully demonstrates how Python, SQLite, Streamlit, and Plotly can be integrated to create a powerful retail analytics solution.

The dashboard enables organizations to monitor sales performance, identify revenue drivers, optimize product portfolios, and support strategic decision-making through interactive visualizations.

Future Enhancements

- Customer Segmentation Analysis.
- Sales Forecasting using Machine Learning.
- Inventory Optimization Models.
- Market Basket Analysis for product association discovery.
- Geographic Sales Analysis by Store Location.
- Real-Time Data Integration from POS Systems.

Overall, the project illustrates how business intelligence dashboards can transform transactional sales data into actionable insights that improve operational efficiency and business performance.